19/02/2021

Code-A

Aakash
Medical | IIT-JEE | Foundations

Regd. Office: Aakash Tower, 8,Pusa Road, New Delhi-110005, Ph.011-47623456

MM : 720

NEET BOOSTER TEST SERIES (NBTS) for NEET-2021

Time : 3.00 Hrs.

Test - 10

Answer Key

1.	(3)	37.	(3)	73.	(3)	109.	(1)	145.	(1)
2.	(2)	38.	(1)	74.	(4)	110.	(1)	146	(2)
3.	(1)	39.	(4)	75.	(1)	111.	(3)	147.	(1)
4.	(2)	40.	(2)	76.	(3)	112.	(4)	148.	(4)
5.	(1)	41.	(1)	77.	(2)	113.	(3)	149.	(3)
6.	(2)	42.	(2)	78.	(1)	114.	(1)	150.	(2)
7.	(4)	43.	(3)	79.	(2)	115.	(3)	151.	(2)
8.	(4)	44.	(3)	80.	(3)	116.	(3)	152.	(1)
9.	(2)	45.	(4)	81.	(4)	117.	(1)	153.	(3)
10.	(3)	46.	(4)	82.	(4)	118.	(4)	154.	(3)
11.	(4)	47.	(1)	83.	(3)	119.	(3)	155.	(4)
12.	(2)	48.	(3)	84.	(1)	120.	(1)	156.	(4)
13.	(3)	49.	(3)	85.	(2)	121.	(3)	157.	(4)
14.	(2)	50.	(2)	86.	(4)	122.	(2)	158.	(1)
15.	(1)	51.	(2)	87.	(2)	123.	(3)	159.	(3)
16.	(3)	52.	(3)	88.	(2)	124.	(2)	160.	(1)
17.	(2)	53.	(4)	89.	(2)	125.	(3)	161.	(2)
18.	(1)	54.	(1)	90.	(3)	126.	(1)	162.	(1)
19.	(3)	55.	(3)	91.	(3)	127.	(4)	163.	(3)
20.	(4)	56.	(3)	92.	(1)	128.	(4)	164.	(4)
21.	(3)	57.	(4)	93.	(3)	129.	(3)	165.	(2)
22.	(1)	58.	(2)	94.	(3)	130.	(3)	166.	(3)
23.	(2)	59.	(2)	95.	(4)	131.	(2)	167.	(1)
24.	(3)	60.	(3)	96.	(1)	132.	(2)	168.	(1)
25.	(3)	61.	(4)	97.	(3)	133.	(3)	169.	(1)
26.	(1)	62.	(1)	98.	(1)	134.	(4)	170.	(1)
27.	(4)	63.	(3)	99.	(3)	135.	(4)	171.	(3)
28.	(3)	64.	(4)	100.	(4)	136.	(2)	172.	(1)
29.	(3)	65.	(2)	101.	(2)	137.	(4)	173.	(3)
30.	(4)	66.	(1)	102.	(3)	138.	(2)	174.	(4)
31.	(2)	67.	(3)	103.	(2)	139.	(2)	175.	(1)
32.	(2)	68.	(1)	104.	(1)	140.	(3)	176.	(1)
33.	(4)	69.	(2)	105.	(3)	141.	(2)	177.	(4)
34.	(4)	70.	(4)	106.	(1)	142.	(2)	178.	(3)
35.	(3)	71.	(3)	107.	(3)	143.	(4)	179.	(3)
36.	(3)	72.	(1)	108.	(3)	144.	(1)	180.	(1)

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MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** Time : 3:00 Hrs.

Test - 10

Hints and Solutions

PHYSICS

1. Answer (3)

$r_n = n^2 r_0 \Rightarrow r_1 = r_0 = r$

$\therefore \quad r_4 = 4^2 \cdot r_1 \Rightarrow r_4 = 16r$

Now, $2\pi(r_4) = 4\lambda$

$\lambda = \dfrac{2\pi(16r)}{4} = 8\pi r$

2. Answer (2)

$E_1 = -13.6$ eV $\Rightarrow PE_1 = -27.2$ eV

$E_2 = -3.4$ eV $\Rightarrow PE_2 = -6.8$ eV

$\Delta PE = PE_2 - PE_1 = [-6.8 - (-27.2)]$ eV

$\Delta PE = 20.4$ eV

3. Answer (1)

$v_n = \dfrac{zv_0}{n}$

$v_2 = \dfrac{zv_0}{2} = v$

$v_4 = \dfrac{zv_0}{4} = \dfrac{1}{2}\left(\dfrac{zv_0}{2}\right) = \dfrac{v}{2}$

4. Answer (2)

$210 \times 0 = 206 \times v - 4v_0$

$v = \dfrac{4v_0}{206}$

5. Answer (1)

Nuclear force is charge independent.

6. Answer (2)

Energy released = $BE_{Product} - BE_{Reactant}$

$= 4c - a - 3b$

7. Answer (4)

1 u = $\dfrac{1}{12}$ (Mass of one C^{12} atom)

1 u is equivalent to energy

$E = 1.66 \times 10^{-27} \times (3 \times 10^8)^2 = 1.49 \times 10^{-10}$ J

$= 149 \times 10^{-12}$ J

or $E = \dfrac{1.49 \times 10^{-10}}{1.6 \times 10^{-19}} = 931.5$ MeV

8. Answer (4)

A free neutron is unstable while inside the nucleus it is stable. Outside nucleus, its mean life is about 1000 s and it decays as

$n \rightarrow p + e + \bar{\nu}$

9. Answer (2)

BE = Δmc^2

= [2(1.0073) + 2(1.0087) – 4.0015] 931.5 MeV

= [4.032 – 4.0015] 931.5 MeV

= 28.4 MeV

10. Answer (3)

$\dfrac{A}{A_0} = \left(\dfrac{1}{2}\right)^{\frac{t}{T_{1/2}}}$

$$\frac{1}{16}=\left(\frac{1}{2}\right)^{\frac{80}{T_{1/2}}}$$

$$T_{1/2}=20 \text{ years}$$

$$T_{mean}=\frac{T_{1/2}}{0.693}=28.9 \text{ years}$$

11. Answer (4)

$${}^{A}_{Z}X \xrightarrow{n\,\alpha} {}^{A-2n}_{Z-n}Y \xrightarrow{n\beta^-} {}^{A-2n}_{Z}W$$

i.e. daughter and parent are isotopes.

12. Answer (2)

$$Q=\left[m_N\left({}^{22}_{11}Na\right)-m_N\left({}^{22}_{10}Ne\right)-m_e\right]c^2$$

$$=\left[m_N\left({}^{22}_{11}Na\right)-11m_e+11m_e-m_N\left({}^{22}_{10}Ne\right)\right.$$

$$\left.-10m_e+10m_e-m_e\right]c^2$$

$$\left[m\left({}^{22}_{11}Na\right)-m\left({}^{22}_{10}Ne\right)-2m_e\right]c^2$$

13. Answer (3)

1 Bq = 1 decay/s

1 Ci = 3.7×10^{10} Bq = 3.7×10^{10} decay/s

14. Answer (2)

Heavy water (D_2O), water, graphite or Beryllium oxide are commonly used as moderator.

15. Answer (1)

For fusion reaction, K.E. is high enough to overcome the coulomb repulsion between nuclei, hence it requires high temperature.

16. Answer (3)

Nuclear density is almost constant for all nuclei.

17. Answer (2)

$\because$ $n_e = n_h$ (For intrinsic semiconductor)

$n_e = 2.5 \times 10^{18}\ m^{-3} = n_h$

Now $n_e + n_h = 2 \times 2.5 \times 10^{18} = 5 \times 10^{18}\ m^{-3}$

$$n' = 5\times10^{18}\times10^{-6} = 5\times10^{12}$$

18. Answer (1)

$\because$ $n_e n_h = n^2$

$2.25 \times 10^{22} \times n_h = (1.5 \times 10^{16})^2$

$2.25 \times 10^{22} \times n_h = 2.25 \times 10^{32}$

$n_h = 10^{10}\ m^{-3}$

19. Answer (3)

For bipolar junction transistor

Length profile $\rightarrow$ C > E > B

Doping profile $\rightarrow$ E > C > B

20. Answer (4)

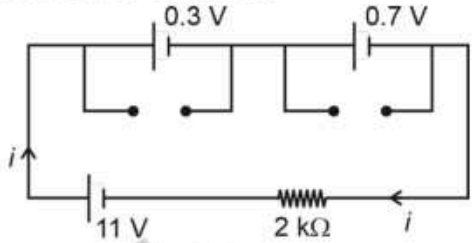
Since knee voltage for Si p-n junction is 0.7 V while for Ge p-n junction it is 0.3 V. Hence only Ge diode will be on and hence $V_A - V_B = 0.3$ V.

21. Answer (3)

$Y = A + B + \bar{A} = 1 + B = 1$

22. Answer (1)

Circuit can be redrawn as



$$i=\frac{(11-0.3-0.7)\text{ V}}{2\text{ k}\Omega}=5\text{ mA}$$

23. Answer (2)

$$v_{rms}=\frac{1}{T}\sqrt{\int_0^T v_0^2\sin^2\omega t dt}=\frac{v_0}{2}$$

24. Answer (3)

Since $I_C + I_B = I_E$

$$\frac{I_E}{I_B}=\frac{I_E}{I_B}+\frac{I_B}{I_B}\Rightarrow Z=Y+1$$

i.e. $Y = Z - 1$

Also $\frac{I_E}{I_C}=\frac{I_C}{I_C}+\frac{I_B}{I_C}$

$$\frac{1}{X}=1+\frac{1}{Y}$$

25. Answer (3)

Transconductance $g_m=\frac{I_0}{V_i}$

$$g_m=\frac{I_0}{I_i R_i}=\frac{\beta}{R_i}$$

$$g_m=\frac{\beta}{2\times10^3}\ \Omega^{-1}$$

$$=\frac{\beta}{2}\times10^{-3}\ \Omega^{-1}$$

26. Answer (1)

$$E_g \le \frac{hc}{\lambda}$$

$$2.4\text{ eV} \le \frac{12400\text{ (eV Å)}}{\lambda\text{ (Å)}}$$

$$\lambda \le \left(\frac{12400}{2.4}\right)\text{Å}$$

$$\lambda \le 5166.67\text{ Å}$$

27. Answer (4)

Here $V_{applied} < V_Z \Rightarrow$ zener diode will not be on, and $I_Z = 0$ and $I_{500\,\Omega} = \frac{5}{1000} = 5\text{ mA}$

28. Answer (3)

If P side is at higher potential than n side, then diode is forward biased.

29. Answer (3)

$Y = \overline{A \cdot B}$ =

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

30. Answer (4)

Depletion width of p-n junction decreases with increase in doping concentration.

31. Answer (2)

$$E_n = -\frac{13.6}{n^2}$$

i.e. $E_n - E_{n-1} = 13.6\left[\frac{1}{(n-1)^2} - \frac{1}{n^2}\right]$

32. Answer (2)

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$E = \frac{h^2}{2m\lambda^2}$$

33. Answer (4)

$$[\text{K.E.}]_{\text{max.}} = \frac{hc}{\lambda} - \phi = \frac{1}{2}mv^2$$

$$\left(\frac{12400}{4000} - 2.5\right)\text{eV} = \frac{1}{2}mv^2$$

$$\frac{2\times(3.1-2.5)1.6\times10^{-19}}{9.1\times10^{-31}} = v^2$$

$$v = 4.6\times10^5\text{ m/s}$$

34. Answer (4)

$$\lambda = \frac{h}{p} \Rightarrow \lambda p = \text{constant}$$

35. Answer (3)

Number of photoelectrons emitted depends on the intensity of light with $(\nu > \nu_0)$.

36. Answer (3)

$$KE_{max} = h\nu - \phi_0$$

$$(KE_{max})_1 = 4.5 - 2.5 = 2\text{eV}$$

$$(KE_{max})_2 = 6.5 - 2.5 = 4\text{eV}$$

$$(KE_{max})_1 : (KE_{max})_2 = 1:2$$

37. Answer (3)

Photoelectric effect shows the particle nature of light.

38. Answer (1)

$$\lambda = \frac{h}{mv} \text{ and}$$

$m_e < m_p < m_n < m_\alpha$

39. Answer (4)

$K_{max} = h\nu - \phi_0$

$K = h\nu - \phi_0$ ... (1)

$K' = 2h\nu - \phi_0$ ... (2)

$K' = 2(K + \phi_0) - \phi_0$

$K' = 2K + \phi_0$

or $K' = K + h\nu$

40. Answer (2)

For proton, $\lambda = \frac{0.286\text{ Å}}{\sqrt{V}}$

$$\lambda = \frac{0.286\text{ Å}}{\sqrt{169}} = 2.2\text{ pm}$$

41. Answer (1)

$$P = \frac{n}{t}(h\nu)$$

$$\frac{n}{t} = \frac{P}{h\nu} = \frac{10000}{6.6\times10^{-34}\times990\times10^3}$$

$$= \frac{10^{33}}{6.6 \times 9.9}$$

$= 1.53 \times 10^{31}$ photons/s

42. Answer (2)

Population inversion occurs in LASER.

43. Answer (3)

$$\frac{1}{\lambda} = R\left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right]$$

For last line of Lyman series

$n_f = 1$ and $n_i = \infty$

$$\frac{1}{\lambda} = R\left[\frac{1}{1} - \frac{1}{\infty}\right] = R$$

$$\frac{1}{\lambda} = 10^{+7}\ m^{-1}$$

44. Answer (3)

$v = \frac{v_0}{n} \Rightarrow$ Speed decreases

$\Rightarrow$ K.E. decreases

P.E. $= -2\left[\frac{13.6}{n^2}\right] \Rightarrow$ P.E. increases with increase in n.

45. Answer (4)

$$E_n = \frac{-13.6(Z^2)}{n^2}$$

$$E_1 = \frac{9}{1}(-13.6) = -122.4\ eV$$

$$E_2 = \frac{9}{4}(-13.6) = -30.6\ eV$$

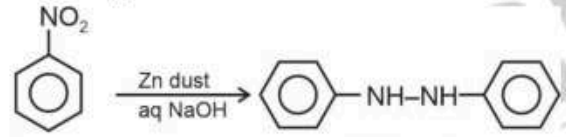
$$E_2 - E_1 = [-30.6 - (-122.4)\ eV = 91.8\ eV$$

CHEMISTRY

46. Answer (4)

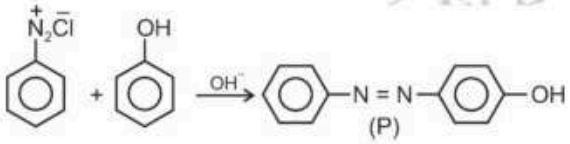
Aliphatic amines are more basic than aromatic amines because in case of aromatic amines, the lone pair of nitrogen is delocalised and is less available for donation.

47. Answer (1)



Nitrobenzene Hydrazobenzene

48. Answer (3)

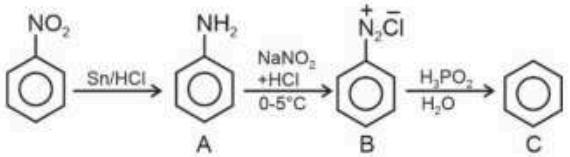


p-hydroxy azobenzene
'orange coloured dye'

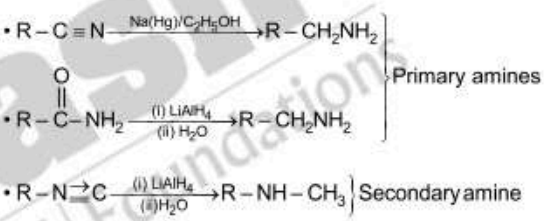
49. Answer (3)

Aromatic diazonium salts are stable at low temperature.

50. Answer (2)



51. Answer (2)



52. Answer (3)

Hinsberg reagent is used to distinguish between 1°, 2° and 3° amines.

53. Answer (4)

Gabriel phthalimide synthesis is used for the preparation of aliphatic primary amines.

54. Answer (1)

Hoffmann bromamide degradation reaction,

$$R - \overset{O}{\overset{\|}{C}} - NH_2 + Br_2 + 4NaOH \rightarrow R\text{–}NH_2 +$$

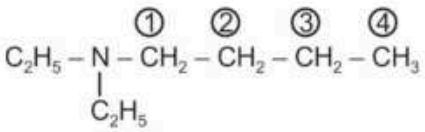
$Na_2CO_3 + 2NaBr + 2H_2O$

The amine so formed contains one carbon less than that present in the amide.

55. Answer (3)

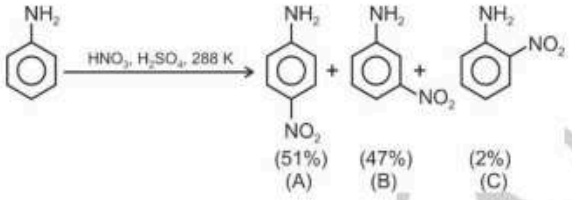
- Higher the intermolecular forces of attraction, higher will be the boiling point.
- The intermolecular hydrogen bonding is more in primary amines than in secondary amines as there are two hydrogen atoms available for hydrogen bond formation in it.

56. Answer (3)



N, N-diethylbutan-1-amine

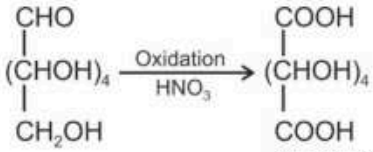
57. Answer (4)



58. Answer (2)

Lactose is composed of β-D-galactose and β-D-glucose

59. Answer (2)



Saccharic acid

60. Answer (3)

In sucrose, the reducing groups of glucose and fructose are involved in glycosidic linkage

∴ It is a non-reducing sugar.

61. Answer (4)

Amylopectin is a branched chain polymer of α-D-glucose units.

62. Answer (1)

Vitamin C is water soluble and vitamin A, D, K are fat soluble.

63. Answer (3)

DNA contains four bases viz. Adenine (A), Guanine (G), Cytosine (C), and Thymine (T).

64. Answer (4)

Amino acid which has more number of amino group than carboxy group are basic in nature.

65. Answer (2)

Rickets (bone deformities in children) in caused by the deficiency of vitamin D.

66. Answer (1)

Ribose contains 5-carbon atoms.

∴ It is a pentose sugar.

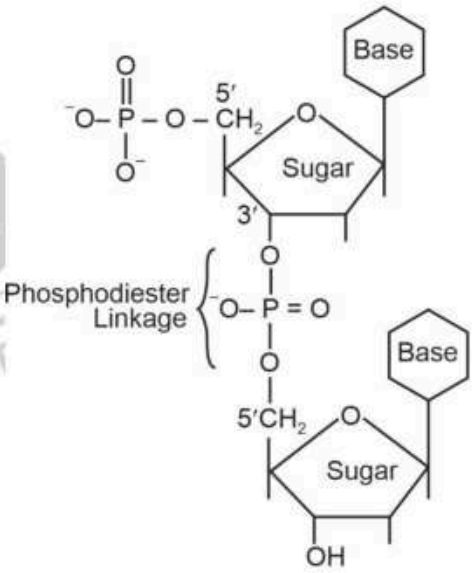
67. Answer (3)

During denaturation of proteins, secondary and tertiary structures are destroyed but primary structure remains intact.

68. Answer (1)

Vitamin A (Retinol) is fat soluble vitamin.

69. Answer (2)



70. Answer (4)

Pyran is 

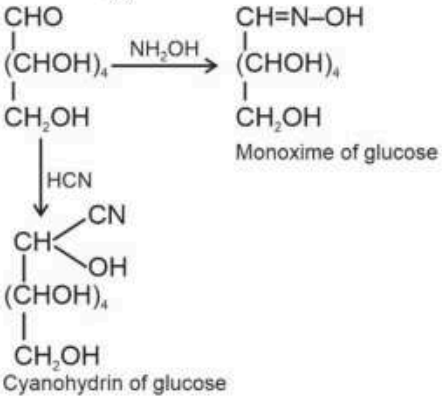
71. Answer (3)

Sucrase decreases the activation energy for the acidic hydrolysis of sucrose.

72. Answer (1)

The two cyclic hemiacetal forms of glucose differ only in the configuration of the hydroxyl group at C1 called anomeric carbon.

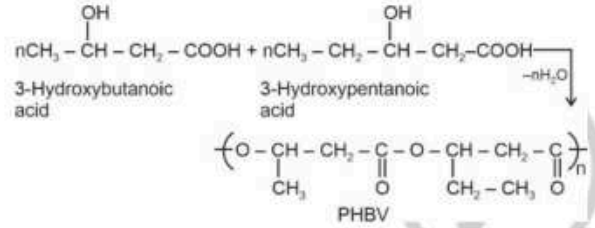
73. Answer (3)



74. Answer (4)

Cellulose is a straight chain polysaccharide composed only of β-D-glucose units which are joined by glycosidic linkage between C-1 of one glucose unit and C-4 of the next glucose unit.

75. Answer (1)



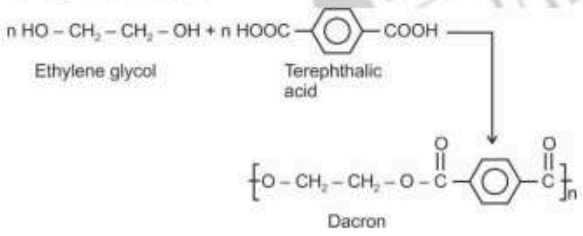
76. Answer (3)

Aliphatic and aromatic primary amines give carbylamine reaction

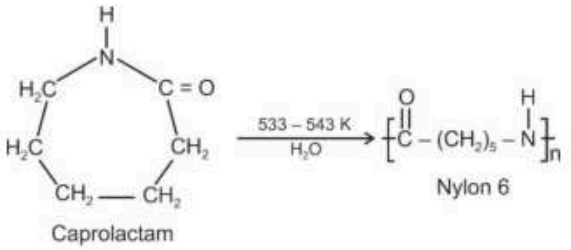
$$R - NH_2 + CHCl_3 + 3KOH \xrightarrow{\Delta} R - NC + 2KCl + 3H_2O$$

77. Answer (2)

Dacron is a polymer of ethylene glycol and terephthalic acid.



78. Answer (1)



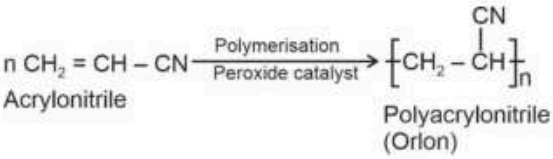
79. Answer (2)

The polymers obtained by addition polymerisation from two different monomers are termed as copolymers.

80. Answer (3)

Glyptal is a polymer of ethylene glycol and phthalic acid.

81. Answer (4)



82. Answer (4)

- Urea-formaldehyde Resin is used for making unbreakable cups and laminated sheets.
- Glyptal is used for the manufacture of paints and lacquers.

83. Answer (3)

Natural Rubber may be considered as a linear polymer of isoprene (2-methyl-1,3-butadiene) and is also called as cis-1,4-polyisoprene.

84. Answer (1)

Ortho-sulphobenzimide is also known as Saccharin, which is the first popular artificial sweetening agent.

85. Answer (2)

BHT (Butylated hydroxy toluene) is an antioxidant whereas sodium benzoate, vegetable oils, table salt are food preservatives.

86. Answer (4)

Penicillin G is a narrow spectrum antibiotic.

87. Answer (2)

Veronal is a tranquilizer.

88. Answer (2)

Norethindrone is an example of synthetic progesterone derivative most widely used as antifertility drug.

89. Answer (2)

- 0.2 percent solution of phenol is an antiseptic.
- 1 percent solution of phenol is a disinfectant.

90. Answer (3)

Cationic detergents are quaternary ammonium salts of amines with acetates, chlorides or bromides as anions.

BOTANY

91. Answer (3)

Crop-fields, garden, aquarium and agro-ecosystem are man-made ecosystems.

92. Answer (1)

In a forest vertical sub-divisions are very clear, top layer has trees, then shrubs, followed by grasses and herbs.

93. Answer (3)

Secondary productivity is the rate of formation of new organic matter by consumers.

94. Answer (3)

Two main limiting factors are light decreases with depth and low nutrient content. Most limiting nutrient of marine ecosystem is nitrogen.

95. Answer (4)

Humus undergoes decomposition at an extremely slow rate. It is highly resistant to microbial action.

96. Answer (1)

Low temperature, anaerobiosis and lignin, chitin rich detritus leads to slower rate of decomposition.

97. Answer (3)

GFC or grazing food chain is the major conduit for energy flow in an aquatic ecosystem.

98. Answer (1)

Standing crop is the mass of living material present at each trophic level at a particular time.

99. Answer (3)

Saprophytes are not given any place in ecological pyramids even though they play a vital role in the ecosystem.

100. Answer (4)

Secondary succession begins in areas that somehow, lost all living organisms that existed there.

101. Answer (2)

The species that invade a bare area are called pioneer species, in xerosere it is lichen and in hydrosere it is phytoplanktons.

The entire sequence of communities that successively change in a given area are called seres.

102. Answer (3)

Primary succession is initiated where no living organism ever existed. Mineralisation is degradation of humus by microbes into inorganic substances.

103. Answer (2)

The reservoir for gaseous type of nutrient cycle, e.g., nitrogen, carbon, exists in the atmosphere.

104. Answer (1)

Gaseous exchanges of phosphorus between organisms and environment are negligible.

105. Answer (3)

Of the various ecosystem services, soil formation accounts for about 50%, recreation and nutrient cycling less than 10%. The cost of climate regulation and habitat for wildlife are about 6 per cent each.

106. Answer (1)

Genetic diversity is a measure of variety in genetic information contained in the organisms.

107. Answer (3)

According to IUCN (2004), total number of plant and animal species described so far is slightly more than 1.5 million. Robert May places the global species diversity at about 7 million.

108. Answer (3)

Out of every 10 animals on this planet, 7 are insects.

So, among animals, insects are the most species-rich taxonomic group.

109. Answer (1)

If we accept May's global estimates, only 22 percent of the total species have been recorded so far.

110. Answer (1)

With very few exceptions, tropics harbour more species than temperate or polar areas. Colombia has about 1400 species of birds.

111. Answer (3)

The tropical Amazonian rain forest in South America has the greatest biodiversity on earth.

112. Answer (4)

Alexander von Humboldt observed that within a region species richness increased with increasing explored area, but only up to a limit.

113. Answer (3)

A stable community should not show too much variation in productivity from year to year.

114. Answer (1)

Key species drives major ecosystem functions. They are most important for the survival of ecosystem.

115. Answer (3)

The 'Sixth Extinction' are estimated to be 100 to 1,000 times faster than in the pre-human times and our activities are responsible for the faster rates.

116. Answer (3)

Lowered resistance to environmental perturbations like drought and increased variability in certain ecosystem processes like productivity, water use etc.

117. Answer (1)

Habitat loss and fragmentation, is the most important cause driving animals and plants to extinction.

118. Answer (4)

Ethical argument relates to what we owe to millions of plant, animal and microbe species with whom we share this planet.

119. Answer (3)

Wild-life safari parks are *ex-situ* conservation strategy.

120. Answer (1)

Convention on biological diversity is called the earth summit. It was held at Rio de Janeiro in 1992 (Brazil).

121. Answer (3)

Electrostatic precipitator can remove over 99% particulate matter from exhaust of thermal power plant.

122. Answer (2)

The air (Prevention and control of pollution) Act came into force in 1981, but was amended in 1987 to include noise as an air-pollutant.

123. Answer (3)

Unburnt hydrocarbons are converted into carbon dioxide and water. Carbon monoxide and nitric oxide are changed into CO_2 and nitrogen gas.

124. Answer (2)

Euro-III stipulates sulphur content at 150 ppm and aromatic hydrocarbon at 42% for petrol. For diesel it is 350 ppm and 42% respectively.

125. Answer (3)

Motor vehicles equipped with catalytic converter should use unleaded petrol because lead in the petrol inactivates the catalyst.

126. Answer (1)

Biomagnification is seen in the aquatic food chain. This phenomenon is well known for mercury and DDT accumulation in organisms.

127. Answer (4)

Natural eutrophication takes thousand of years while accelerated eutrophication is a fast process.

128. Answer (4)

Human excreta can be recycled into a resource as natural fertiliser, in Ecosan toilets.

129. Answer (3)

Polyblend are fine powder of recycled modified plastic. This is mixed with bitumen that is used to lay roads.

130. Answer (3)

Open-burning dumps are not really much of a solution as the amount of garbage generation in metros has increased due to same.

131. Answer (2)

Relative contribution of CO_2 is about 60%, CH_4 (20%), CFCs(14%) and N_2O(6%) to global warming.

132. Answer (2)

Montreal Protocol was signed at Montreal (Canada) in 1987 and became effective in 1989, to control the emission of ozone depleting substances.

133. Answer (3)

NFC(1988) recommended 33% forest cover for plains and 67% for the hills.

134. Answer (4)

Joint Forest Management was introduced in the 1980s, to work closely with the local communities for protecting and managing forests.

135. Answer (4)

Bad ozone is a deadly pollutant and powerful pollutant.

ZOOLOGY

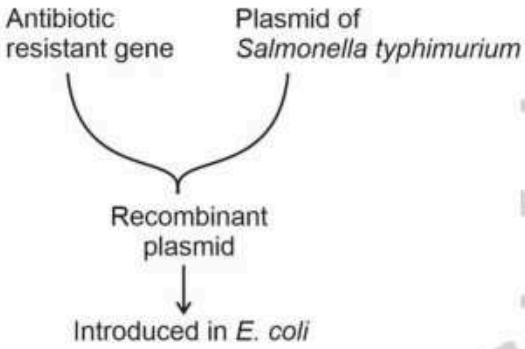
136. Answer (2)

Herbert Boyer observed that restriction enzymes cut the DNA in a particular fashion, which leaves sticky ends on the DNA strands

137. Answer (4)

All are part of biotechnology.

138. Answer (2)



139. Answer (2)

The first type II restriction endonuclease is *Hind* II isolated from the bacterium *Haemophilus influenzae*.

140. Answer (3)

Restriction endonuclease bind to the DNA and cut each of the two strands of the double helix at specific points in their sugar phosphate backbones.

141. Answer (2)

Sticky ends form hydrogen bonds with their complementary cut counterparts. This stickiness of the ends facilitates the action of the enzyme DNA ligase.

142. Answer (2)

Restriction enzymes cut at specific sequences so the same restriction enzyme is used to cut the source DNA and vector because it will produce fragments with the same complementary sticky ends, facilitate the bonds formation between them.

143. Answer (4)

DNA fragments are negatively charged and move towards anode according to their size. Smaller the fragment size, farther it moves.

144. Answer (1)

The most commonly used matrix in gel electrophoresis is agarose which is a natural polymer extracted from sea weeds.

145. Answer (1)

The separated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution.

146 Answer (2)

Size of vector should be small.

147. Answer (1)

In pBR322, recognition sequence of *Eco*RI is not present in the amp^R or tet^R gene.

148. Answer (4)

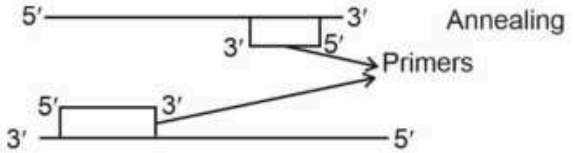
Number of copies of DNA = 2^n

n = No. of cycles

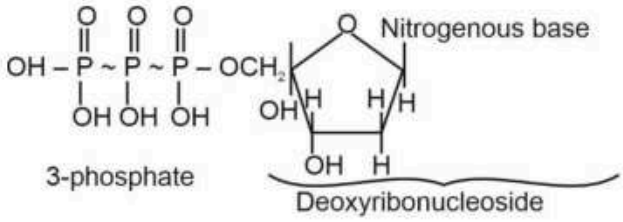
149. Answer (3)

The phosphodiester bond links a 3′ carbon of one sugar to the 5′ carbon of another sugar.

150. Answer (2)



151. Answer (2)



152. Answer (1)

Use of the thermostable *Taq* polymerase enables running the PCR at high temperature. It is obtained from *Thermus aquaticus*.

153. Answer (3)

Sparger in sparged stirred tank bioreactor, creates bubbles which increases the surface area for O_2 transfer.

154. Answer (3)

Large scale culture of host cells in bioreactors is a part of upstream processing.

155. Answer (4)

dNTPs are used in PCR.

156. Answer (4)

All of them have capability to replicate in bacteria.

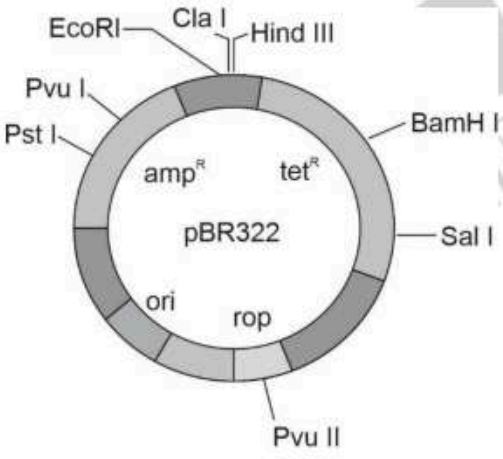
157. Answer (4)

Retroviruses in animals have the ability to transform normal cells into cancerous cells. Disarmed retroviruses are used to deliver desirable genes into animal cells.

158. Answer (1)

Non-transformants being sensitive to ampicillin will not grow on ampicillin containing medium.

159. Answer (3)



160. Answer (1)

Isolation of donor DNA

↓

Fragmentation by restriction endonuclease

↓

Isolation of desired DNA fragment

↓

Ligation of DNA fragment into a vector

↓

Transferring the recombinant DNA into the host

↓

Culturing the host cells

161. Answer (2)

If any protein encoding gene is expressed in a heterologous host, it is called a recombinant protein.

162. Answer (1)

Early diagnosis of disease is possible by the techniques like PCR, ELISA and recombinant DNA technology.

163. Answer (3)

Desired gene is introduced in selectable marker gene.

164. Answer (4)

(1)	Ori	–	Replication starts from ori sequence
(2)	rop	–	Codes for the proteins involved in the replication of the plasmid
(3)	Palindromic nucleotide sequence	–	Base pairs that reads same on the two strands when orientation of reading is kept the same

165. Answer (2)

In *Eco*RI, R is derived from the name of strain RY13.

166. Answer (3)

Bioremediation is the use of micro-organisms for the removal of pollutants. e.g., *Pseudomonas putida* used for clearing oil spills.

167. Answer (1)

Cotton bollworm belongs to order Lepidoptera.

168. Answer (1)

cry gene is named so because it codes for crystalline protein.

169. Answer (1)

(1)	Green revolution	–	Use of agrochemicals (chemical fertilisers and pesticides)
(2)	Bt toxin	–	Create pores in the epithelial cells of midgut of insects
(3)	RNAi	–	Present in all eukaryotes as a method of cellular defense

170. Answer (1)

Dicer enzymes act on dsRNA.

171. Answer (3)

At present, about 30 recombinant therapeutics have been approved for human use the world over.

172. Answer (1)

Eli Lilly an American company prepared two DNA sequences corresponding to A and B chains.

173. Answer (3)

ADA deficiency can be cured by bone marrow transplantation.

174. Answer (4)

Rosie cow was developed in the year 1997.

175. Answer (1)

In 1997, an American company got patent rights on Basmati rice through the US patent and trademark office.

176. Answer (1)

Indian Basmati rice was crossed with semi-dwarf variety of rice.

177. Answer (4)

Ethics means a set of standards by which a community regulates its behaviour.

178. Answer (3)

For chemical safety testing, transgenic animals are made more sensitive to toxic substances.

179. Answer (3)

Meloidogyne incognitia causes root knot in tobacco plant.

180. Answer (1)

Bt toxin is not secreted to outside, instead, present inside the bacterium as inclusion body.

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